σ-Additive Probability on Orthomodular Lattices: A Survey and an Open Problem

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Abstract

The passage from a finitely coherent assignment of probabilities to a countably additive measure is, in the Boolean case, governed by a single condition (countable additivity) and executed by classical machinery (Carathéodory; Loomis–Sikorski; Stone duality). On a *non-distributive* orthomodular lattice (OML) — the algebra of propositions of a quantum system — this machinery breaks, and the question of when finite coherence forces countable behaviour is open. We survey the problem at the level needed to begin work on it. After fixing the apparatus (orthomodular lattices, the gap between orthogonality and meet-zero, the state/meet-additive/charge ladder, the McDonald–Bimbó dual space), we separate the question into two axes. The *extension* axis is classical (Horn–Tarski feasibility) and fully settled: on $L(H)$ no state of any kind extends, by an elementary finite construction. The *descent* axis is where the genuinely open question lives, and we locate it precisely. The combinatorial route — seeking the object among concrete, σ-orthocomplete, set-representable OMLs — is **settled and does not produce it**: the interleaving property that would make σ-additivity exceed finite additivity on a concrete non-Boolean lattice is satisfied trivially by the plain product $\prod_n \mathrm{MO}_2$, which is concrete, σ-complete, non-Boolean, infinite — and exactly the *segregated* object (non-distributivity confined to finite blocks) that a relational probability theory must exclude. The deeper reason is that a faithful σ-tribe representation has a distributive image, forcing the OML Boolean, so no relational measure can arise from a faithful descent to points; the forcing boundary to Boolean is *subadditivity* of the separating states (Pták–Pulmannová), not σ-additivity. Read "relational" as *no hidden realisation* — the measure not recoverable from a posited auxiliary space of definite states — which, by Fine's theorem, is exactly a *contextual* state: one not in the closed convex hull of the lattice's dispersion-free states. The open problem is then sharp: **does a non-Boolean, σ-complete, concrete OML carry a σ-additive contextual state whose contextuality is σ-essential** (witnessed by no finite sub-OML)? The finite version is *already settled* — Wright's pentagon is a finite concrete OML with a contextual state — so the live content is the σ-essential refinement, which is a *compactness failure* of exactly the Paper I (CE) type and rejoins the programme's origin. The state of the question: well-posed and principled but *uninhabited* — no witness ($\prod_n \mathrm{MO}_2$ fails it; Wright is finite), no impossibility proof. A positive answer is the non-Boolean analogue of localic measure theory; a negative one a σ-essential impossibility theorem, reinstating empiricist-underdetermination as a theorem. We review what is known and ruled out, and close with the two exits.

1 Introduction

All measurement is finite, yet the frameworks that organise empirical knowledge — probability theory, quantum mechanics — rest on infinite structures. The passage from finite observation to infinite structure is not automatic; it requires a commitment no finite body of evidence can compel. In probability theory the locus of that commitment is *countable additivity*:

$$A_1 \supseteq A_2 \supseteq \cdots, \quad \bigcap_n A_n = \varnothing \quad \Longrightarrow \quad \mu(A_n) \to 0.$$

De Finetti [8] held that only finite additivity is empirically grounded; Kolmogorov [24] adopted σ-additivity as an axiom.

The same tension recurs once observations need not be jointly performable. An observation *context* — a set of measurements performable together — is Boolean; gluing contexts along shared observations (a categorical colimit) returns:

$$\text{contexts mutually compatible} \Rightarrow \text{Boolean algebra}, \qquad \text{incompatible} \Rightarrow \text{non-distributive OML}$$

(Gunji et al. [17]; the dichotomy organises the companion paper [25]). The OML is *forced* by the context structure, not postulated. Incompatibility — quantum complementarity, but equally contextual experimental design or interfering sensors — drives the algebra out of the Boolean world. One caveat fixes the scope: for non-quantum examples the incompatibility must be *operationally imposed* (contexts genuinely non-co-realizable, no joint probability space); it cannot arise intrinsically from one classical system's dynamics, since any finite family of classical observables on a common space jointly distributes (Kolmogorov), hence stays Boolean — contextuality there is failure of a global section (Abramsky–Brandenburger [1]; Fine [12]). The closed subspaces of a Hilbert space H form the prototype $L(H)$, not the premise.

On a non-distributive OML, Carathéodory's outer measure — needing a Boolean algebra of sets — does not apply, and the rescue is local to $L(H)$:

- For $L(H)$, Gleason's rigidity theorem [15] rescues it; Hilbert space is special and no general-OML analogue is known.
- Non-distributivity *per se* does not remove the 2-valued homomorphisms the Boolean engine needs — MO_3 is non-distributive yet carries them (§2.1, §4) — but it removes the *guarantee*; for $L(H)$, $\dim \geq 3$ they fail outright (Kochen–Specker [23]).

Whether enough survive is a question of concreteness, not non-distributivity — where an OML is *concrete* (Definition 2.7) when its elements can be faithfully modelled as honest *sets of outcomes*, equivalently when it carries an order-determining set of two-valued states (Gudder [16]); "enough survive" is then almost the definition. This is the qualifier on which the open problem turns (§4, §5).

The organising split. This note asks the shared question — *when does finite coherence force countable behaviour?* — on an OML. The vague question splits into two axes that the Boolean case fuses, and the survey is built around the split:

Axis	**Asks**	**Status**
Extension	does a state extend to a charge on $S_0(A)$?	settled (on $L(H)$, *none* does)
Descent	does the charge concentrate on physical points?	**open**

The relational standpoint. Two established moves motivate the approach:

- De Finetti [8] — probability without a presupposed sample space: coherent commitments about events, not measures on a given Ω.
- Birkhoff–von Neumann — propositions as a lattice (the OML of §2), not as subsets of a phase space.

Combining them, we start from the propositions and their entailment order and ask what that relational structure alone determines — the measure-theoretic counterpart of pointless (localic) topology. The dual is then not a space of points but the McDonald–Bimbó space of *filters* (§2.3), and a positive answer to the open problem would be a *relational* σ-additive probability theory on a non-distributive lattice — the non-Boolean analogue of localic measure theory. (The §5 statement makes "relational" precise as a *contextual* state, no hidden realisation; developed further in [26].)

2 Preliminaries

We collect the lattice-theoretic apparatus; all notions are standard.

2.1 Orthomodular lattices, and the orthogonal/meet-zero gap

Definition 2.1 (Orthocomplemented lattice)**.** A *bounded lattice* is a partially ordered set in which every pair a, b has a join $a \vee b$ and meet $a \wedge b$, with greatest element $\mathbf{1}$ and least element $\mathbf{0}$. An *orthocomplementation* is a map $a \mapsto a^\perp$ satisfying, for all a, b:

(i) $a \wedge a^\perp = \mathbf{0}$ and $a \vee a^\perp = \mathbf{1}$ (complement),
(ii) $a^{\perp\perp} = a$ (involution),
(iii) $a \leq b \Rightarrow b^\perp \leq a^\perp$ (order-reversing).

Definition 2.2 (Incomparable; meet-zero; orthogonal)**.** For elements a, b of an orthocomplemented lattice, three relations of increasing strength:

- *incomparable*: $a \not\leq b$ and $b \not\leq a$ — neither entails the other (an order condition);
- *meet-zero*: $a \wedge b = \mathbf{0}$ — no common refinement;
- *orthogonal*, written $a \perp b$: $a \leq b^\perp$ (equivalently $b \leq a^\perp$) — one entails the other's negation, *decisively incompatible*.

Each implies the one above it ($a \perp b \Rightarrow a \wedge b = \mathbf{0} \Rightarrow$ incomparable, for $a, b \notin \{\mathbf{0}, \mathbf{1}\}$). For two *distinct atoms*, incomparable and meet-zero coincide — an atom has only $\mathbf{0}$ below it, so $a \wedge b \in \{\mathbf{0}, a\}$, and $a \wedge b = a$ would force $a \leq b$; orthogonality remains strictly stronger.

Remark 2.3 (Meet-zero versus orthogonal)*.* The implication $a \perp b \Rightarrow a \wedge b = \mathbf{0}$ reverses in every Boolean algebra, but fails once the lattice is non-distributive: there meet-zero is strictly weaker than orthogonality. The distinctions developed below all turn on this separation.

Definition 2.4 (Orthomodular lattice)**.** An orthocomplemented lattice A is *orthomodular* if

$$a \leq b \quad \Longrightarrow \quad b = a \vee (a^\perp \wedge b). \tag{1}$$

A *Boolean algebra* is the special case where $\wedge$ distributes over $\vee$; every Boolean algebra is orthomodular, not conversely. The motivating non-Boolean example is $L(H)$, the closed subspaces of a Hilbert space with $a^\perp$ the orthogonal complement.

Definition 2.5 (Completeness notions)**.** Two countable-completeness conditions on an OML A are distinguished throughout, the second strictly weaker. Here $\bigvee_n a_n$ denotes the *join* (least upper bound) of $\{a_n\}$ in the order of A, unique when it exists; "$\exists \bigvee_n a_n \in A$" asserts that it does.

- A is *σ-complete* if
$$\forall \{a_n\}_{n\in\mathbb{N}} \subseteq A : \quad \exists \textstyle\bigvee_n a_n \in A$$
(equivalently every countable family has a meet, by $\bigwedge_n a_n = (\bigvee_n a_n^\perp)^\perp$);
- A is *σ-orthocomplete* if
$$\forall \{a_n\}_{n\in\mathbb{N}} \subseteq A \ \big(a_i \perp a_j \text{ for } i \neq j\big) : \quad \exists \textstyle\bigvee_n a_n \in A.$$

The orthogonal families are a subclass of all countable families, so σ-completeness quantifies over a strictly larger domain and is the stronger hypothesis:

$$\{a_n\} \text{ pairwise orthogonal } \subseteq \{a_n\} \text{ arbitrary} \qquad \Longrightarrow \qquad \sigma\text{-complete} \Rightarrow \sigma\text{-orthocomplete}.$$

Only the weaker condition is needed below. For a state s, σ-additivity is the requirement

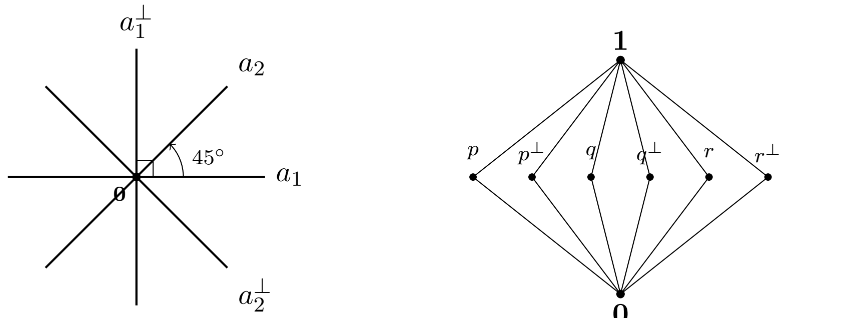
$$s\Big(\textstyle\bigvee_n a_n\Big) = \sum_n s(a_n) \qquad \text{for every countable } \{a_n\} \text{ with } a_i \perp a_j \ (i \neq j),$$

whose left side invokes $\bigvee_n a_n$ *only* for orthogonal families — so σ-orthocompleteness is exactly the hypothesis that makes it well-posed. Full σ-completeness is the stronger condition manipulated by the Loomis–Sikorski representation (§2.4).

Example 2.6 (MO$_n$, and the gap made concrete)**.** MO$_n$ is the lattice with $\mathbf{0}$, $\mathbf{1}$, and n complementary pairs of incomparable atoms, all sharing only $\mathbf{0}$ and $\mathbf{1}$. The geometric model is n lines through the origin: each line a is orthogonal to its perpendicular $a^\perp$, while two atoms from distinct pairs are non-perpendicular lines. It is orthomodular and non-distributive ($n \geq 2$): two atoms drawn from *distinct* complementary pairs have

$$a \wedge b = \mathbf{0} \quad \text{(meet only at the origin)} \qquad \text{yet} \qquad a \not\perp b,$$

so meet-zero $\neq$ orthogonal already here. MO$_3$ is the smallest non-distributive OML.



$a_1 \perp a_1^\perp$ (right angle),
$a_1 \not\perp a_2$ but $a_1 \wedge a_2 = \mathbf{0}$

complementary pairs $(p, p^\perp), (q, q^\perp), (r, r^\perp)$;
any two non-paired atoms meet at $\mathbf{0}$, join at $\mathbf{1}$

Figure 1: MO$_n$ illustrated. *Left:* the geometric model for MO$_2$ — four atoms as lines through the origin in $\mathbb{R}^2$. The pair $a_1, a_1^\perp$ meets at a right angle (orthogonal), whereas a_1, a_2 meet at 45°: their lattice meet is still $\mathbf{0}$ (only the origin in common), yet they are *not* orthogonal — the meet-zero $\neq$ orthogonal gap. *Right:* the Hasse diagram of MO$_3$, the smallest non-distributive OML: three complementary pairs of incomparable atoms between $\mathbf{0}$ and $\mathbf{1}$.

Definition 2.7 (OMP; order-determining states; concrete logic)**.** An *orthomodular poset* (OMP) weakens Definition 2.4: joins $a \vee b$ are required only for *orthogonal* pairs. A set S of states on A is *order-determining* if it separates the order, i.e.

$$\big(s(a) \leq s(b) \text{ for all } s \in S\big) \implies a \leq b.$$

An OMP is a *concrete logic* (equivalently *set-representable*) if it embeds into $(\mathcal{P}(X), \subseteq, X \setminus (-))$ for some set X, preserving existing orthogonal joins — i.e. its propositions are modelled as actual *sets of outcomes*. Concreteness is equivalent to admitting an order-determining set of *two-valued* states (Gudder [16]), and it is the property that separates the open frontier from the settled cases (§5).

Remark 2.8 (Concreteness does not close the gap)*.* A set-representable OMP needs only closure under complement and *disjoint* union [4]; intersection-closure would force a field of sets, hence Boolean. The lattice meet $a \wedge b$ is the greatest L-member inside $A \cap B$, so

$$\underbrace{A \cap B = \varnothing}_{\text{orthogonal}} \implies \underbrace{a \wedge b = \mathbf{0}}_{\text{meet-zero}}, \qquad \text{but not conversely.}$$

Concreteness does not force the converse: MO$_3$ is itself concrete (Definition 2.7; Gudder [16]), so the paradigmatic meet-zero-$\neq$-orthogonal example is already a concrete logic. What closes the gap is a richness/atomicity condition, not concreteness:

$$a \wedge b = \mathbf{0} \Rightarrow a \perp b \quad \iff \quad \text{every nonempty } A \cap B \text{ contains a nonzero } L\text{-member},$$

the abstract form being Tkadlec's *Boolean orthoposet* condition [35] (which does *not* mean Boolean *algebra*).

2.2 The state/meet-additive/charge ladder

Definition 2.9 (State; meet-additive state; charge-extendible)**.** Let A be an OML. A *state* is a map $s : A \to [0,1]$ with $s(\mathbf{1}) = 1$ that is additive on *orthogonal* pairs: $s(a \vee b) = s(a) + s(b)$ whenever $a \perp b$; that is, a probability assignment whose additivity is guaranteed only on decisively incompatible propositions. Three strengthenings, differing in which pairs additivity is demanded on and strictly increasing in general (Example 2.10 on MO_3):

(1) *State*: $s(a \vee b) = s(a) + s(b)$ whenever $a \perp b$.
(2) *Meet-additive*: $s(a \vee b) = s(a) + s(b)$ whenever $a \wedge b = \mathbf{0}$ (binary, strictly stronger — it reaches the gap pairs of Definition 2.2). We avoid the word "valuation" here: in lattice theory it denotes the modular function $v(a)+v(b) = v(a\wedge b)+v(a\vee b)$, a different condition.
(3) *Charge-extendible*: $\sum_i s(a_i) \le 1$ for every pairwise-meet-zero family $\{a_i\}$ ($a_i \wedge a_j = \mathbf{0}$, $i \neq j$; strictly stronger again). This is the condition necessary for extension to a charge on the dual space (Definition 2.13) and the *extension* condition of §3.

Example 2.10 (The ladder is strict)**.** On MO_3 with complementary pairs $(p, p^\perp), (q, q^\perp), (r, r^\perp)$, the maximally mixed $s \equiv \frac{1}{2}$ is meet-additive (clears (2)) yet fails (3) at $\{p, q, r\}$: $\frac{1}{2} + \frac{1}{2} + \frac{1}{2} = \frac{3}{2} > 1$. So no state on MO_3 is charge-extendible.

Definition 2.11 (Normal and singular states on $L(H)$)**.** On $A = L(H)$ a state s is

- *normal* if it is completely additive on every orthogonal family of projections,

$$s\Big(\bigvee_{i\in I} p_i\Big) = \sum_{i\in I} s(p_i) \quad \text{for all (not only countable) orthogonal } \{p_i\},$$

  equivalently $s(\cdot) = \mathrm{tr}(\rho\,\cdot)$ for a density operator ρ (Gleason [15], dim$\geq$3);
- *singular* (purely) if

$$s(e) = 0 \quad \text{for every finite-rank projection } e.$$

Every state decomposes into a normal and a singular part (Takesaki); the two classes are exactly those distinguished by the arguments of §3.

2.3 The McDonald–Bimbó dual space

Definition 2.12 (Dual space; representation map)**.** The McDonald–Bimbó duality [27] assigns to an OML A a compact space

$$S_0(A) = \big(F(A), \subseteq, \perp_A, P(A), T\big),$$

where $F(A)$ is the set of *filters* of A, ordered by $\subseteq$ and carrying an orthogonality relation $\perp_A$ and a Stone topology T, and $P(A) = \{\uparrow p : p \in A\} \subseteq F(A)$, with $\uparrow p = \{a \in A : a \geq p\}$, is the set of *principal* filters — the "physical points," the realisation datum (canonical, unlike the non-canonical pure points of the Boolean case). For $a \in A$ set $h(a) = \{x \in F(A) : a \in x\}$; then h is an OML isomorphism onto the $\perp$-stable clopens $\mathrm{CO}(S_0(A))^\dagger$. Define $\mu(h(a)) = s(a)$. Here $h(a)$ is the set of points at which a holds and μ is the state transported onto those sets; the open problem is whether μ is preserved under countable operations on them.

Definition 2.13 (Charge)**.** A *charge* on $S_0(A)$ is a map $\nu : \mathrm{CO}(S_0(A)) \to [0,1]$ on the *full* Boolean algebra of clopen subsets (not merely the $\perp$-stable ones in the image of h), with

$$\nu(S_0(A)) = 1, \qquad \nu(U \cup V) = \nu(U) + \nu(V) \text{ whenever } U \cap V = \varnothing.$$

A state s *extends to a charge* if some such ν satisfies $\nu(h(a)) = s(a)$ for all $a \in A$. Because h preserves meets, a pairwise-meet-zero family in A maps to disjoint clopens, so a charge necessarily satisfies $\sum_i s(a_i) \leq 1$ over such families — the charge-extendibility condition of Definition 2.9(3).

Remark 2.14 (The duality is finitary). The representation map h satisfies

$$h\Big(\bigvee_{i\leq n} a_i\Big) = \Big(\bigcup_{i\leq n} h(a_i)\Big)^{\perp\perp} \ \text{(finite)}, \qquad h\Big(\bigvee_{n} a_n\Big) \neq \Big(\bigcup_{n} h(a_n)\Big)^{\perp\perp} \ \text{(countable)} :$$

finitary OML homomorphisms need not preserve infinite suprema. This is the central obstruction the open problem of §5 confronts, and it is self-dual: since $a \mapsto a^\perp$ is an order anti-isomorphism, $\bigwedge_n a_n = (\bigvee_n a_n^\perp)^\perp$, so the failure for countable meets is the De Morgan dual of the failure for countable joins. The rival duality of Cannon–Döring [5] is also finitary and carries no measure; we use McDonald–Bimbó because it carries $P(A)$ as explicit structure.

2.4 The Boolean engine: Stone duality

In the Boolean case the descent equivalence

$$\sigma\text{-additive} \quad \Longleftrightarrow \quad \text{the measure concentrates on the physical points}$$

is proved through *Stone duality*, in four steps [26]:

(S1) a Boolean event algebra B embeds in the clopen algebra $\mathrm{CO}(\mathrm{St}\, B)$ of its Stone space;
(S2) a charge on B extends to a Baire measure on $\mathrm{St}\, B$ (Carathéodory);
(S3) σ-additivity $\Longleftrightarrow$ the measure vanishes on meagre Baire sets (Rao–Rao [33]);
(S4) meagre-null $\Longleftrightarrow$ the measure lives on the ultrafilters closed under countable intersection = the realised points.

Steps (S3)–(S4) are the measure-theoretic face of the *Loomis–Sikorski* representation,

$$\sigma\text{-complete Boolean algebra} \ \cong \ (\sigma\text{-tribe of sets}) \,/\, (\sigma\text{-ideal}),$$

the countable-join bookkeeping that drives the equivalence. The contrast with the OML case is exact:

Boolean	**OML**
Stone duality + Loomis–Sikorski (the engine)	**no analogue** (§4.3)

Supplying such an engine, or proving none exists, is the open problem of §5.

Remark 2.15 (Why the bridge breaks: meet-closed but not join-prime). A "realised" point x carries two conditions, which Rao–Rao's "vanishes on meagre sets" makes coincide in the Boolean case:

(M) closed under countable meets; (J) $x \ni \bigvee_n a_n \Rightarrow x \ni a_n$ some n (off the join-defect).

Non-distributivity separates them. For a principal filter $x = {\uparrow}p$:

(M) always holds.
(J) can fail: $p \leq \bigvee_n a_n$ does not force $p \leq a_n$.

In $L(H)$ this is witnessed by $a_n = \mathrm{span}(e_n)$ and $p = \mathrm{span}(e_1 + e_2)$:

$$p \ \leq \ \textstyle\bigvee_n a_n = H, \qquad \text{yet} \qquad p \ \not\leq \ a_n \ \text{ for every } n.$$

So $\uparrow p$ is a realised point lying inside the join-defect $D = h(\bigvee_n a_n) \setminus \bigcup_n h(a_n)$. Topologically $\uparrow p$ is *isolated* in $S_0(L(H))$, hence:

$$D \text{ non-meagre (OML)} \qquad \text{vs.} \qquad D \text{ nowhere dense (Boolean).}$$

This is the finitary obstruction of Remark 2.14 on the dual: the category route (clopens modulo the meagre ideal) fails on the same D as the measure route, and reduces to the MacNeille completion of §4.3(ii).

3 Two axes: extension and descent

The central question — *when does a state s on an OML A extend to a σ-additive measure on $S_0(A)$?* — splits into two **independent** axes that the Boolean case fuses. A state s on A faces:

	Extension axis	**Descent axis**
Question	Does μ extend to a finitely additive charge on the *full* Boolean clopen algebra of $S_0(A)$?	Once extended, does the measure concentrate on the physical points $P(A)$?
Governed by	the meet-zero/orthogonal gap (§2.1)	σ-additivity — needs the missing engine (§2.4)
Status	settled: Horn–Tarski / Pitowsky feasibility; finite case = decidable LP; can fail for every state.	open.

The extension axis is the abstract form of a familiar question: extending a finitely-additive assignment on incompatible propositions to a single joint distribution is exactly the feasibility problem behind the Bell inequalities (Fine [12]: a Bell model exists iff the correlations admit a joint distribution; Pitowsky [30]: the Bell inequalities are the facets of a correlation polytope), and Bell and Kochen–Specker non-classicality are jointly an extension problem for partial Boolean algebras (Budroni–Morchio [2], with the necessary-and-sufficient conditions of Horn–Tarski [21]). What is distinctive here is only that the extension is sought against the dual $S_0(A)$ rather than within a fixed scenario; the obstruction is the same finitary one, and it is settled. Two observations pin it down.

Proposition 3.1 (Finite case)**.** *For finite A the extension condition is a decidable linear program (Horn–Tarski [21] / Pitowsky correlation-polytope feasibility [30]; Budroni–Morchio [2] for the partial-Boolean-algebra formulation); since h preserves meets, meet-zero elements map to disjoint clopens and a charge must satisfy $\sum_i s(a_i) \leq 1$ over pairwise-meet-zero families. It can fail for* every *state (*MO_3*, Example 2.10).*

Proposition 3.2 (Infinite $L(H)$, all states)**.** *Let* $\dim H = \infty$. No *state on $L(H)$ — normal or singular — extends to a charge on $S_0(L(H))$.*

Proof. The argument embeds a copy of MO_2 in $L(H)$ whose four atoms are infinite-dimensional, so that the obstruction reaches the singular states as well. Write $H = H_0 \otimes \mathbb{C}^2$ with H_0 infinite-dimensional, and fix an orthonormal basis e, f of $\mathbb{C}^2$. Set

$$a_1 = H_0 \otimes \mathbb{C}e, \quad a_1^\perp = H_0 \otimes \mathbb{C}f, \quad a_2 = H_0 \otimes \mathbb{C}(e{+}f), \quad a_2^\perp = H_0 \otimes \mathbb{C}(e{-}f).$$

These four infinite-dimensional subspaces form a copy of MO_2 in $L(H)$:

$$a_1 \perp a_1^\perp, \quad a_2 \perp a_2^\perp, \quad a_i \vee a_i^\perp = H; \qquad a_i \wedge a_j = \mathbf{0} \;\text{ but }\; a_i \not\perp a_j \;\text{ (cross pairs).}$$

Orthoadditivity with $s(\mathbf{1}) = 1$ gives $s(a_i) + s(a_i^\perp) = 1$ for $i = 1, 2$, using only additivity on orthogonal pairs, so it holds for *every* state. The family $\{a_1, a_1^\perp, a_2, a_2^\perp\}$ is pairwise meet-zero, and since h preserves meets ($h(a \wedge b) = h(a) \cap h(b)$), its members map to disjoint clopens; hence any charge satisfies $\sum_i s(a_i) \le 1$. The two counts collide:

$$\sum_i s(a_i) = 2 \quad \text{(orthoadditivity)} \qquad \text{versus} \qquad \sum_i s(a_i) \le 1 \quad \text{(any charge)}.$$

The construction is finite ($n = 4$); σ-completeness of the lattice plays no role. □

The MO_2 count is standard (orthoadditivity together with $s(\mathbf{1}) = 1$; see [22]) and $\sum s(a_i) = 2$ is already state-independent in $L(\mathbb{C}^2)$. The role of the *infinite-dimensional* realisation is to extend the obstruction to the singular states ($s(e) = 0$ on every finite-rank e):

Atoms of MO_2	$s(a_i) + s(a_i^\perp)$	**Reaches**
lines in $L(\mathbb{C}^2)$ (finite rank)	$= 1$ only if s charges finite rank	normal states only
$H_0 \otimes (\cdot)$, $\dim H_0 = \infty$	$= 1$ regardless of finite-rank behaviour	**all states, incl. singular**

Keeping each atom infinite-dimensional drops the finite-rank precondition of the line argument (Remark 3.3), and that is what closes the extension axis on $L(H)$ completely.

Remark 3.3 (An alternative argument for normal states only). A second, technique-disjoint argument reaches the normal states without the MO_2 family. If $s(e) > 0$ on some finite-dimensional e, it is positive on a 2-plane e_0; taking k pairwise-non-orthogonal lines $p_1, \ldots, p_k \subset e_0$ yields $2k$ pairwise-meet-zero lines with $s(p_i) + s(p_i^\perp) = s(e_0)$, so a charge forces

$$k\, s(e_0) \le 1 \ \text{ for all } k \quad \Longrightarrow \quad s(e_0) = 0,$$

contradicting $s(e_0) > 0$. Lifting s to a functional via Mackey–Gleason / Bunce–Wright [3] ($B(H)$ has no type I_2 summand) and applying the Takesaki normal/singular decomposition identifies which states the argument reaches:

reached	nonzero normal part ($s(e) > 0$ on some finite e); includes every normal (Gleason) state
missed	purely singular ($s(e) = 0$ for all finite e); e.g. ultrafilter vector states, Calkin-algebra pullbacks

Proposition 3.2 subsumes the missed case; this argument is recorded because its technique is independent and locates the singular states precisely.

4 What is known

The extension axis is settled — on $L(H)$ no state extends (Proposition 3.2); the descent axis turns on whether the Boolean engine of §2.4 has an OML analogue. Three things bear on that:

1. the Boolean benchmark such an analogue must match (§4.1);
2. the partial OML results, positive and obstructive (§4.2);
3. the routes already known to be blocked (§4.3).

4.1 The Boolean benchmark

The conditions an OML engine would have to reproduce. The Kelley–Vladimirov–Pták characterisation (Fremlin [13], Thm 391D) is complete: a Boolean algebra carries a strictly positive σ-additive measure iff it satisfies all three conditions below — each of which loses its meaning off the distributive world.

Boolean (KVP) condition	**OML analogue**
Dedekind σ-completeness	exists (σ-orthocompleteness)
weak (σ,∞)-distributivity	**none** — relies on $\bigwedge$ distributing over $\bigvee$
chargeability	**none** — no structural characterisation on OMLs

4.2 Positive and obstructive OML results

Positive (require structure rich enough to approximate Hilbert-space geometry):

- Gleason [15] — $L(H)$, $\dim \geq 3$; σ-additivity *on the lattice*.
- Bunce–Wright [3] — JBW projection lattices; Chetcuti–Dvurečenskij [6].

Obstructive (Boolean machinery cannot be transplanted):

- Pták–Pulmannová [32] — a *finitary* condition (*subadditive*-unitality $s(a \vee b) \leq s(a) + s(b)$, no countable joins) already collapses the *lattice* to Boolean (their Thm 1); the OM*poset* version fails (Müller's set-representable counterexample [28]).
- Navara [29] — regularity does not force σ-additivity. Pták [31] — exotic state spaces via Greechie pasting.

4.3 Three routes to a σ-OML engine, all unavailable

The descent argument needs an OML analogue of the Boolean engine — a σ-Stone duality, or equivalently a Loomis–Sikorski representation. The three known routes are all unavailable, but not on the same footing: routes (i) and (ii) are closed by *theorem* (the hypothesis that would yield the representation provably forces Boolean, resp. leaves the category), whereas route (iii) is open in the strong sense — no construction is known, and no impossibility theorem rules one out (Remark 4.1). It is route (iii), restated, that is the Open Problem of §5.

(i) **RDP / effect-algebra.** Loomis–Sikorski holds for σ-MV algebras [10] and monotone σ-effect algebras *with* the Riesz Decomposition Property [11]; but a lattice effect algebra has RDP iff it is MV, and OML $\cap$ MV = Boolean (Kalmbach [22]), so the hypothesis that would yield the representation collapses the class:

$$\mathrm{OML} + \mathrm{RDP} \iff \text{Boolean}.$$

The *sharp-skeleton* evasion — representing the OML as the *sharp* elements of an ambient RDP effect algebra, so that only the ambient carries RDP — also closes: in any RDP effect algebra the sharp elements coincide with the center and are hence Boolean (Jenča 2001, Cor. 4.3; via Greechie–Foulis–Pulmannová 1995), so a non-Boolean sharp skeleton witnesses RDP-failure. Route (i) is closed against this refinement too (σ-completeness is not on the load path). The MacNeille completion of an OML need not be orthomodular (Harding [18]); one cannot complete to absorb countable joins. The topological version of this route — clopens of $S_0(A)$ modulo the meagre ideal — is this same completion in another form (the regular-open algebra), and collapses non-distributivity (Remark 2.15).

(ii) **σ-Stone duality.** None is *known*, and none is *ruled out*: the existing dualities are finitary (McDonald–Bimbó, Remark 2.14) or equational with no set/tribe representation

(Freytes [14], on σ-complete OMLs), but no theorem forbids a countable-join-preserving one.

4.4 Prior art on point-free quantum probability

Every existing point-free or directed-context construction fails at least one of the two requirements — σ-additivity and a natively non-distributive structure — as tabulated below.

Gleason tradition
Varadarajan [36], Kalmbach [22], Pták–Pulmannová [32]: σ-additive on a non-distributive OML, but real-valued on a fixed lattice — not point-free.

Döring 2009 [9] point-free on the spectral presheaf over a directed poset — but provably only *finitely additive*, on a distributive Heyting algebra.

Bohrification (HLS)
[19, 20] point-free and the internal valuation *is* σ-additive (Scott-continuous) — but factors through an internally *distributive* locale, σ-additive only per commutative context. Non-distributivity is tamed, not carried natively.

Localic valuations
Vickers [37], Simpson [34], Coquand–Spitters [7]: point-free σ-additive, but on *frames* — distributive by definition.

OML Stone dualities
McDonald–Bimbó [27] and Cannon–Döring [5]: purely structural, no measure, no σ-version.

Each entry satisfies some of the requirements but not all. The open case is their conjunction, satisfied by no existing construction:

$$\boxed{\text{point-free} \times \sigma\text{-additive} \times \text{natively non-distributive} \times \text{directed-system-built.}}$$

(The directed-context-yields-non-distributivity construction is not new — it is the shape of Bohrification and of colimit-over-Boolean-contexts generation. Novelty would lie only in the σ-additive $\times$ point-free $\times$ native combination.)

Remark 4.1 (The boundary of the negative results). Two qualifications delimit what is ruled out.

(i) **No impossibility theorem covers the concrete class.** The known no-go results are finite or two-valued, and the one positive occupant is non-concrete:

$$L(H)\ (\dim \geq 3): \quad \text{no two-valued states at all (Kochen–Specker [23]),}$$

a fortiori none order-determining. Concreteness is the qualifier on which the open frontier turns.

(ii) **No σ-Loomis–Sikorski exists**, and this is visible in the dualities: both take infinite joins as a closure, not a union,

$$\text{Cannon–Döring: } \mathrm{cls}(\textstyle\bigcup S_i); \qquad \text{McDonald–Bimbó: } (\textstyle\bigcup h(a_n))^{\perp\perp}.$$

This is not to be overstated: Gudder's concrete logics [16] *are* set-representable non-Boolean orthoposets — but as point-ful posets, not via a σ-Loomis–Sikorski theorem.

4.5 What the topos route does and does not buy

The topos / Bohrification programme is the one machinery that supplies a genuinely point-free *and* σ-additive (directed-continuous) valuation on a quantum logic, so it is the natural candidate to crack route (iii). A mechanism-level read of the two principal constructions shows it is *the same wall, with tools*: the σ-additive valuation always lives on a *distributive* object, and the bridge to the non-distributive lattice routes its countable additivity through per-context Boolean blocks. The non-distributivity is relocated into base-poset variation and a coarse-graining map, never carried by the measure. This is the topos analogue of the universal floor (§5.1): a measure that meets the non-distributive structure faithfully forces it Boolean.

Bohrification (Heunen–Landsman–Spitters). A state on A becomes a *continuous probability valuation* $\mu : O(\Sigma(A)) \to [0,1]_l$ on the internal Gelfand spectrum $\Sigma(A)$, which is an internal *locale* — by construction a compact regular *frame* [20, §2, §6], hence internally distributive. The defining continuity axiom $\mu(\bigvee_i U_i) = \bigvee_i \mu(U_i)$ for directed families (Def. 6.11) is precisely the localic, Scott-continuous form of σ-additivity — but it is asked of $O(\Sigma(A))$, a frame, where $\bigvee$ is distributive. The non-commutativity of A sits entirely in the base poset $\mathcal{C}(A)$ of commutative subalgebras and in the topos's intuitionistic internal logic; the valuation never integrates over a non-distributive lattice.

Remark 4.2 (The HLS valuation does not reach the prize — objection and answer). Theorem 6.19 of [20] is a bijection between continuous probability valuations on $\Sigma(A)$ *and* probability measures on $\mathrm{Proj}(A)$ — the genuinely non-distributive object — so it *looks* like a σ-additive measure on a non-distributive OML. It is not a counterexample to the wall, for two independent reasons.

Primary (σ-essential).
By Definition 6.17(a) a probability measure on the countably complete OML $\mathrm{Proj}(A)$ is a map that *restricts to a σ-Boolean-algebra morphism on every countably complete Boolean sublattice*. Its countable additivity is thus *defined block-by-block on the Boolean sub-structure* and glued by naturality over $\mathcal{C}(A)$; it never crosses a non-distributive (non-commuting) join. This is the *antithesis* of σ-essential contextuality (§5.1): the prize demands contextuality witnessed by the countable whole and by *no* Boolean sub-structure, whereas the topos route confines *all* of its countable additivity *inside* Boolean contexts. This argument is independent of concreteness.

Secondary (non-concrete).
The non-distributive object here is $\mathrm{Proj}(A)$, the projection lattice of a (Rickart) C^*-algebra — the Gleason / $L(H)$-type lattice, non-concrete by Kochen–Specker (Rmk 4.1). So Bohrification *recasts the Gleason occupant* (the top row of §4.4) in topos language; it does not reach the concrete class on which the open frontier turns.

Döring–Isham (spectral presheaf). Measures live on the clopen subobjects $\mathrm{Sub}_{cl}(\Sigma)$, a complete Heyting algebra / locale — distributive, and *explicitly not* a σ-algebra [9]. The non-distributive $\mathrm{Proj}(H)$ enters only through *daseinisation* $\delta : \mathrm{Proj}(H) \to \mathrm{Sub}_{cl}(\Sigma)$, which is a coarse-graining, not a lattice homomorphism: outer δ^o preserves all joins but not meets, inner δ^i preserves all meets but not joins, and — in Wolters' words [38] — "it cannot preserve both, as $[\mathrm{Sub}_{cl}(\Sigma)]$ is distributive, whereas $[\mathrm{Proj}(H)]$ is nondistributive." This is the exact topos twin of the finitary-duality wall (Rmk 2.14). The measure is in general only *finitely* additive; the ceiling is driven by the state class (to cover type-III / KMS / non-normal states), and σ-additivity is recovered only *locally* — countable additivity for families pairwise orthogonal at a single context V, equivalently within one Boolean block, and only for normal states. As with

HLS, the apparent "σ-additive on projections" statement holds only over pairwise-*orthogonal* (hence commuting, hence single-context, hence Boolean) families: the additivity never crosses a non-distributive join.

The f.a.-vs-σ diagnosis. The route lands on the programme's finitely-additive-versus-σ gap exactly: σ-additivity is available *precisely where the object is distributive* — the internal frame, or the per-context Boolean blocks where Loomis–Sikorski already applies ("the first arrow is free") — and degrades to finite additivity (Döring) precisely when a single global measure is demanded across the non-distributive whole. Of the four requirements (point-free $\times$ σ-additive $\times$ *natively non-distributive* $\times$ directed-system-built), the topos route satisfies three and misses exactly the one that is the crux: native non-distributivity. The machinery engages the problem; it does not hold a handle on it. It *sidesteps* the wall (an internally distributive Gelfand spectrum is a design goal, so constructive Gelfand duality goes through) rather than *settling* the σ-essential question — which stays open, neither inhabited nor closed.

5 The open problem

Open Problem. Is there a non-Boolean, σ-complete, *concrete* OML carrying a σ-additive *contextual* state — a state with no global hidden joint distribution, equivalently one not lying in the closed convex hull of the lattice's dispersion-free (two-valued) states — whose contextuality is *σ-essential*, i.e. witnessed by no finite sub-OML? Or can one prove that no such state exists?

This is the relational-probability prize made precise. "Relational" means *no hidden realisation*: the measure must not be recoverable from positing an auxiliary space of definite states behind the propositions. Three point-spaces must be kept apart (the recurring trap of this subject): the *canonical* dual $P(A) = \{\uparrow p\}$ (one principal filter per *element*, §2.3); the *dispersion-free states* (two-valued states, one per consistent global *valuation* — the hidden-variable points); and the *external (A)-points* (Gleason's Hilbert rays, Remark 5.2). The canonical $P(A)$ is *permitted*; the objection is only to building the measure on a *smuggled* space — the (A)-points, or equivalently a global hidden-variable joint over the dispersion-free states. A state has a hidden realisation exactly when it is a mixture of dispersion-free states (Fine's theorem), so a relational measure is precisely a *contextual* state, and the reduction below turns the existence question into this one. (Note this is a condition on the dispersion-free states, *not* on $P(A)$: a witness is point-rich on $P(A)$ yet has no global dispersion-free joint — the two spaces are different, which is exactly why "permitted" and "escaped" do not conflict.) (We do not pursue the stricter "strictly point-free / no canonical points either" reading: it makes concreteness itself the enemy and collapses against the universal that follows, so it offers no inhabitable target.)

5.1 The reduction to contextuality, and what frames it

The universal floor. A *faithful set representation* (σ-tribe / Loomis–Sikorski, preserving $\wedge, \vee$ as $\cap, \cup$) has a distributive image, so it forces the OML Boolean (Remark 2.8). Hence no non-Boolean OML admits one, and a relational measure can never arise from a faithful descent to points. What a concrete OML *does* have is an order-determining family of dispersion-free states; the live question is whether every state is built from them.

The reduction. A σ-additive state w on a concrete OML is *non-contextual* iff it lies in the closed convex hull of the dispersion-free states iff it admits a global hidden joint. So:

$$\text{relational (no hidden realisation)} \iff \text{contextual (not spanned by dispersion-free states).}$$

The handle is *spanning*, not simplicity — non-uniqueness of the decomposition is irrelevant; only the *existence* of a dispersion-free decomposition matters. The prize is therefore a concrete σ-complete OML with a σ-additive state outside the dispersion-free hull.

Why σ-essential, and why it is principled. The finite version of this is *already inhabited*: Wright's pentagon [39] is a finite, concrete OML (a Greechie loop of length five, hence a lattice) with a separating family of dispersion-free states yet a state outside their hull — a contextual state. So "concrete OML with a contextual state" is a 1978 fact, and the general spanning theorem is correspondingly false. What Wright does not supply is contextuality that *requires* the countable structure: the pentagon's is witnessed by a finite sub-OML (itself). The prize must therefore be *σ-essential* — contextual in the σ-complete whole but in no finite sub-OML. This is not a device to evade Wright: "every finite piece has a global section, the countable whole has none" is a *compactness failure*, the same finite-additivity-has-compactness / σ-additivity-lacks-it structure that is the core of Paper I (CE). The descent question thereby rejoins the programme's origin.

Status. The σ-essential cell has *no known inhabitant* and *no impossibility proof.* $\prod_n \mathrm{MO}_2$ provably fails it (its contextual states are only finitely additive — diffuse states on its central $P(\mathbb{N})$; σ-additivity forces concentration on points and kills them, §5.2); Wright is finite; no construction is in hand. The question is well-posed and principled but uninhabited. (Computations and the literature verdict: companion notes `reading1_prize_reduction.md` and `direction2_gate_finding.md`.)

5.2 What the combinatorial route settles, and why it is not enough

It is tempting to look for the object among concrete σ-orthocomplete OMLs, on the grounds that a state's σ-additivity invokes $\bigvee_n a_n$ only for orthogonal families (Definition 2.5). This route is **settled, and it does not produce the object.** The property that would make σ-additivity genuinely exceed finite additivity on a concrete non-Boolean lattice is the interleaving condition

$$(\bigstar)\quad \exists\, p,\ \exists\,\{a_n\}\ \text{infinite orthogonal},\quad p \wedge a_n = 0\ (\forall n),\quad p \not\leq a_n\ (\forall n), \tag{2}$$

and $(\bigstar)$ is satisfied *trivially* by the plain product $\prod_n \mathrm{MO}_2$: take $a_n = a$ in coordinate n (zero elsewhere), and $p = \bigvee_n b_n$ where $b_n = b$ in coordinate n (zero elsewhere) — a genuine countable *orthogonal* join, so $p = (b, b, b, \dots)$ exists by σ-orthocompleteness. Then $p \wedge a_n = 0$ (coordinate n: $a \wedge b = 0$ in MO_2; elsewhere $a_n = 0$) and $p \not\leq a_n$ (coordinate n: $b \not\leq a^\perp$). Since $\prod_n \mathrm{MO}_2$ is concrete, σ-complete (a product of finite, hence complete, lattices), non-Boolean, and infinite, it satisfies $(\bigstar)$ and *every* combinatorial hypothesis one might impose — yet it is exactly the *segregated* object the open problem must exclude. So the combinatorial axioms (σ-completeness, concreteness, $(\bigstar)$) are *not* the discriminator; the missing ingredient is non-segregation, which is a point-free notion, not a closure property.

Remark 5.1 (The witness has no points, and the Boolean boundary). The universal closure above (a faithful set representation forces distributivity) already rules out the point-based route for every non-Boolean OML. The witness $\prod_n \mathrm{MO}_2$ shows the failure concretely and at the level of points: MO_2 admits *no* two-valued *homomorphism* — each of its four separating two-valued *states* ($a \mapsto 0$, $b \mapsto 0$, etc.) fails $\omega(a \vee b) \leq \omega(a) + \omega(b)$ since $a \vee b = \mathbf{1}$, so none preserves $\vee$ — and $\prod_n \mathrm{MO}_2$ inherits this, since its diagonal copy $\{\mathbf{0}, \mathbf{1}, (a, a, \dots), (a^\perp, \dots), (b, b, \dots), (b^\perp, \dots)\}$ is a sub-MO_2 on which any homomorphism would restrict to one of MO_2's (none exists). (This is special to MO_2 and its products — a generic non-Boolean OML such as $\mathbf{2} \times \mathrm{MO}_2$ *does* carry homomorphisms, e.g. the projection; the universal obstruction is the distributivity argument, not homomorphism-scarcity.) So the witness is concrete (Definition 2.7: order-determining two-valued *states*) and σ-complete yet has no points in the homomorphism sense at all — the point-free target is forced, not engineered. The same arithmetic records the sharp Boolean boundary: an OML is Boolean iff it carries a unital set of *subadditive* states (Pták–Pulmannová [32]), so concrete non-Boolean OMLs exist precisely *because* their separating two-valued states fail subadditivity — and the forcing property is subadditivity, not σ-additivity.

A worked instance fixes the picture. Navara's construction [29] produces an infinite, σ-orthocomplete, rich OML whose inner block is a finite *stateless* Greechie logic — which is exactly what makes Navara's $\mathcal{L}$ non-concrete. Replacing the stateless block by MO_2 (write $\mathcal{L}_2$) removes the only source of non-concreteness: the assembly is a product of a sublogic of a Kalmbach horizontal sum (blocks glued only at $\{0, 1\}$, *not* an atom-sharing Greechie loop), and pure horizontal sums of concrete logics are concrete, as are sublogics and products. So $\mathcal{L}_2$ is concrete, σ-orthocomplete, non-Boolean, and carries ($\bigstar$) — yet it is not a new object: it is $\prod_n \mathrm{MO}_2$ with extra scaffolding, segregated in the same way, and it likewise admits no σ-tribe representation. It is the cautionary example showing that exhibiting ($\bigstar$) and concreteness together is easy and does *not* reach the open problem.

The art constrains the open (σ-complete) problem from both sides without closing it: *no* impossibility theorem covers the concrete class (Remark 4.1), while the two *theorem-closed* routes to an engine — RDP and MacNeille (§4.3(i),(ii)) — are ruled out, leaving only the third, σ-Stone duality, which is itself open. What is required is not a new idea in the abstract but a representation in that gap: non-distributive enough to evade routes (i),(ii), yet σ-faithful where the known dualities are merely finitary.

Remark 5.2 (Why $L(H)$+Gleason does *not* already settle this — the (A)/(B) point-space equivocation). "Probability from relations, point-free" may *look* delivered by $L(H)$+Gleason. It is not: the appearance rests on conflating two point-spaces, and *realisation* is defined as concentration on (B).

	Point-space	**Gleason's relation to it**
(A)	Hilbert rays of H	*requires* them — ρ in $s = \mathrm{tr}(\rho\,\cdot)$ is built from rays
(B)	MB dual filters $P(A)$	*removes* them, but only by being a representation theorem

Gleason is thus the *most* point-ful result available (every lattice state $\mapsto$ the point datum ρ). What $L(H)$ exhibits is a *separation*, not point-freeness:

on the lattice	σ-additive measures exist (Gleason, normal states);
on the dual $S_0(L(H))$	no state extends to a charge (Prop. 3.2),
	so a fortiori none concentrates on $P(A)$.

It cannot serve as the existence proof required, so the point-free σ-additive structure of §5 is the only candidate engine.

6 Directions

With the extension axis closed on $L(H)$ (Proposition 3.2), the combinatorial route shown to deliver only segregated objects (§5.2), and the prize reduced to a σ-essential contextual state (§5.1), the problem has a clean win/kill dichotomy. Two exits, each a genuine terminus. *(Pen-and-paper kit:* `problemset_oml_descent.{tex,pdf}` — a self-contained working problem-set with the apparatus, the formal target, and these exits as concrete sub-tasks.*)*

Exhibit a σ-essential witness

Construct an infinite concrete OML carrying a σ-additive contextual state that no finite sub-OML witnesses — the relational prize. The likely attack is a compactness-failure construction in the CE mould: a concrete σ-complete OML whose finite sublogics are all classically interpretable (dispersion-free states span them), but whose countable joins force a Kochen–Specker-type obstruction only in the limit. The faithful set representation is ruled out at every stage (§5.2), so the obstruction must be carried by the σ-structure itself, not by any finite block. *Target sharpened (CE-routing subsession, 2026-06-12): the witness must put its infinitary structure* ***off-center*** *(irreducible / non-central-infinite) — a product of finite blocks like $\prod_n \mathrm{MO}_2$ is excluded a priori, since it forces all infinitary content into the Boolean center where the Stone argument kills it. Pasting / countable colimit of Wright-type blocks is the place to look.*

Prove no such witness exists

A compactness/spanning theorem: every σ-additive state on a σ-complete concrete OML lies in the closed convex hull of its dispersion-free states *except* where a finite sub-OML is already contextual (Wright). This is a Type-5 impossibility, and it would reinstate the *empiricist-underdetermination* conclusion as a theorem: relational σ-additive probability that is σ-essential cannot exist, so the relational content the programme seeks is operationally invisible after all. The Boolean boundary — subadditivity of the separating states (Pták–Pulmannová, Remark 5.1) — is the natural lever. *Caveat (2026-06-12): this cannot reuse the $\prod_n \mathrm{MO}_2$ / Stone-over-center mechanism — vacuous off-center — so a genuinely different finite-witnessing argument is needed.*

A candidate direction now closed: the singular case for $L(H)$. One might have hoped a singular state would extend where no normal state does, providing the first explicit "extends but may fail to concentrate" example; Proposition 3.2 rules this out, catching every state, singular included. The reduction is recorded because the singular case was expected to require infinitary methods and does not (Remark 6.1).

Remark 6.1 (Finite versus countable: finite suffices). Is the obstruction to extending a singular state realised by a *finite* or only by a *countable* pairwise-meet-zero family? The two arguments divide on exactly this point:

line argument (Rmk 3.3)	finitary, but **vacuous** on singular ($s(e) = 0\ \forall$ finite e)
MO_2**, ∞-dim atoms (Prop. 3.2)**	finite ($n = 4$), $\sum_i s(a_i) = 2$ state-independently

So the singular obstruction is *not* entangled with the descent-axis passage, as the vacuity of the line argument might suggest: it is realised by a finite family, hence a self-contained finitary problem, separable from the general construction and not governed by the finitary-to-σ passage.

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